Pressure—Temperature Dependence of the Water—Graphene Interfacial Tension, with Insights into Cavitation under Negative Pressure

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P(Mpa)	$T(K)$	γ_{WG} (mN/m)
-100	320	79
0.1	320	88.4
10	320	100.8
100	320	125.8
-100	360	54.9
0.1	360	84.2
10	360	90.4
100	360	122.2

Tab. S1: Water-graphene surface tension (γ_{WG}), as fonction of the temperature an the pressure.

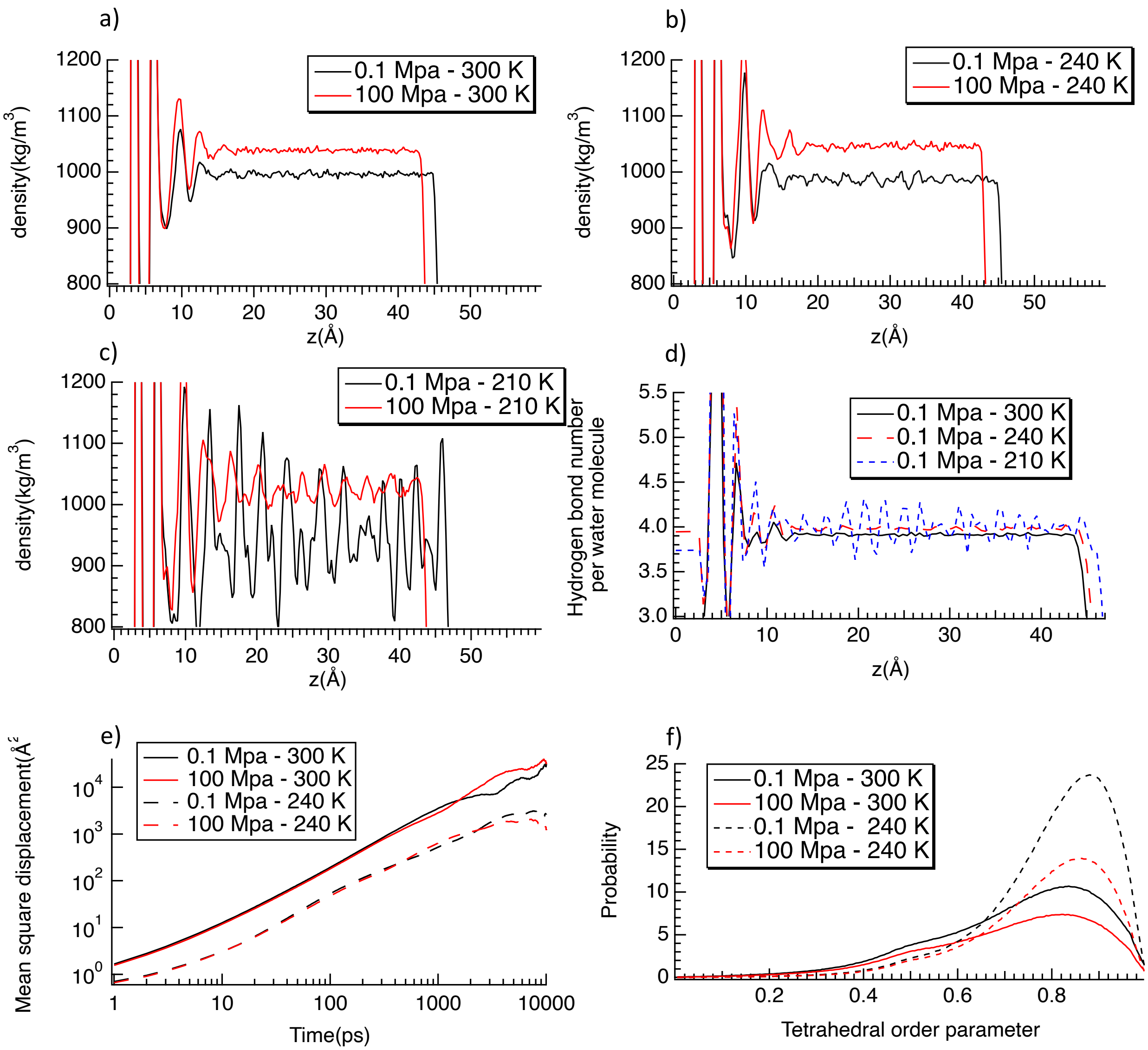


FIG. S1: Density profiles of the water center of mass along the z direction at 0.1 MPa and 100 MPa for (a) 300 K, (b) 240 K, and (c) 210 K. (d) z-dependent hydrogen-bond number per molecule (HBn) at 0.1 MPa for 300 K, 240 K, and 210 K. HBn is computed using the geometric criteria of Luzar and Chandler[1]. (e) Mean-square displacement and (f) distribution of the tetrahedral order parameter at 240 K and 300 K for both 0.1 MPa and 100 MPa. The tetrahedral order parameter quantifies the local structural order of the first coordination shell[2]: values close to 0 correspond to disordered configurations, whereas values close to 1 indicate a nearly perfect tetrahedral arrangement.

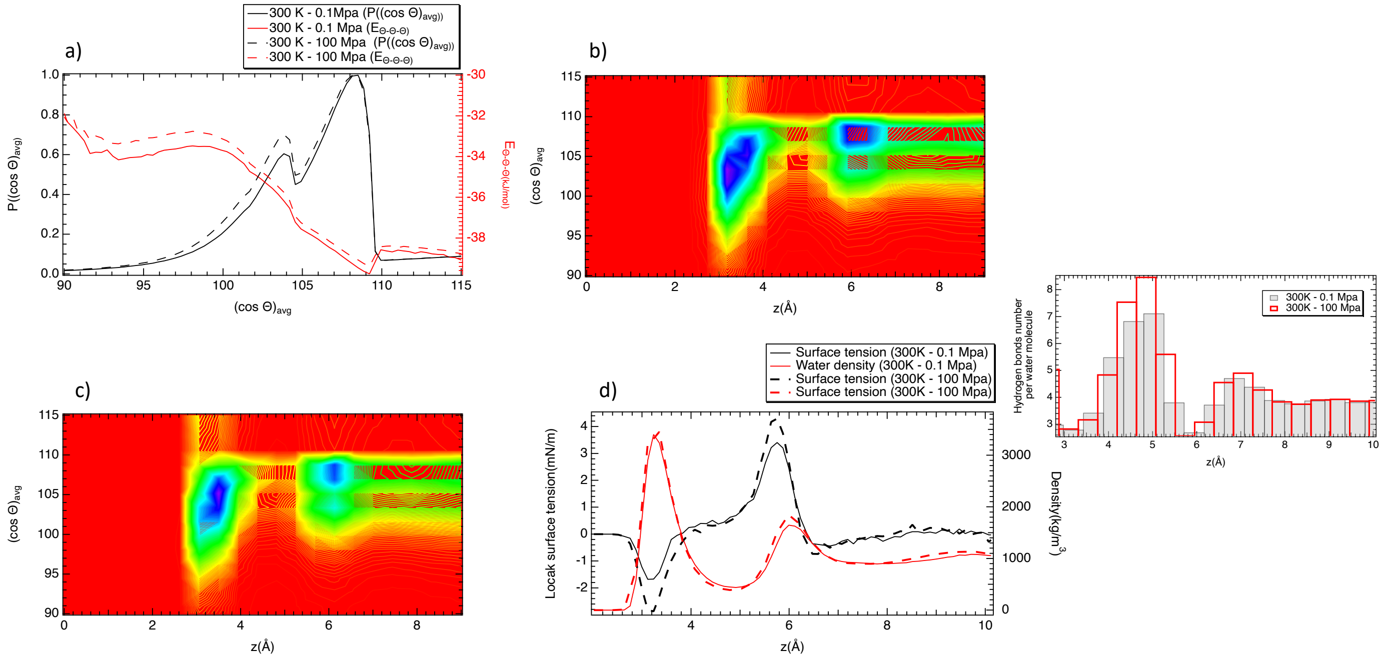


FIG. S2: (a) Probability distribution $(\cos\theta)_{\mathrm{avg}}$ of the average angular order parameter (left axis) and the corresponding angular interaction energy $\mathrm{E}_{\theta-\theta-\theta}$ (right axis). (b,c) Two–dimensional maps of $(\cos\theta)_{\mathrm{avg}}$ as a function of the position z across the confined region, highlighting the spatial heterogeneity of the orientational structure at 0.1 Mpa (b) and 100 Mpa (c). (d) Local surface tension profile (left axis) along z together with the water density profile (right axis); the inset shows the distribution of hydrogen-bond coordination per water molecule along z direction.

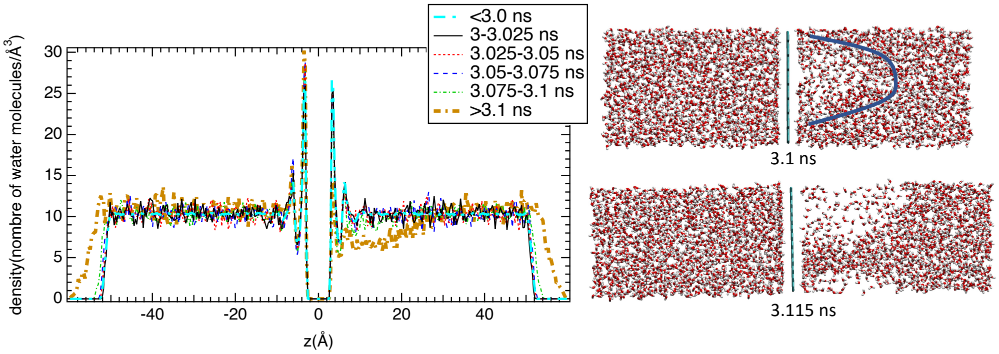


FIG. S3: Density profile of the water center of mass along the z direction at 280 K and -200 MPa, together with representative snapshots illustrating the onset of cavitation.

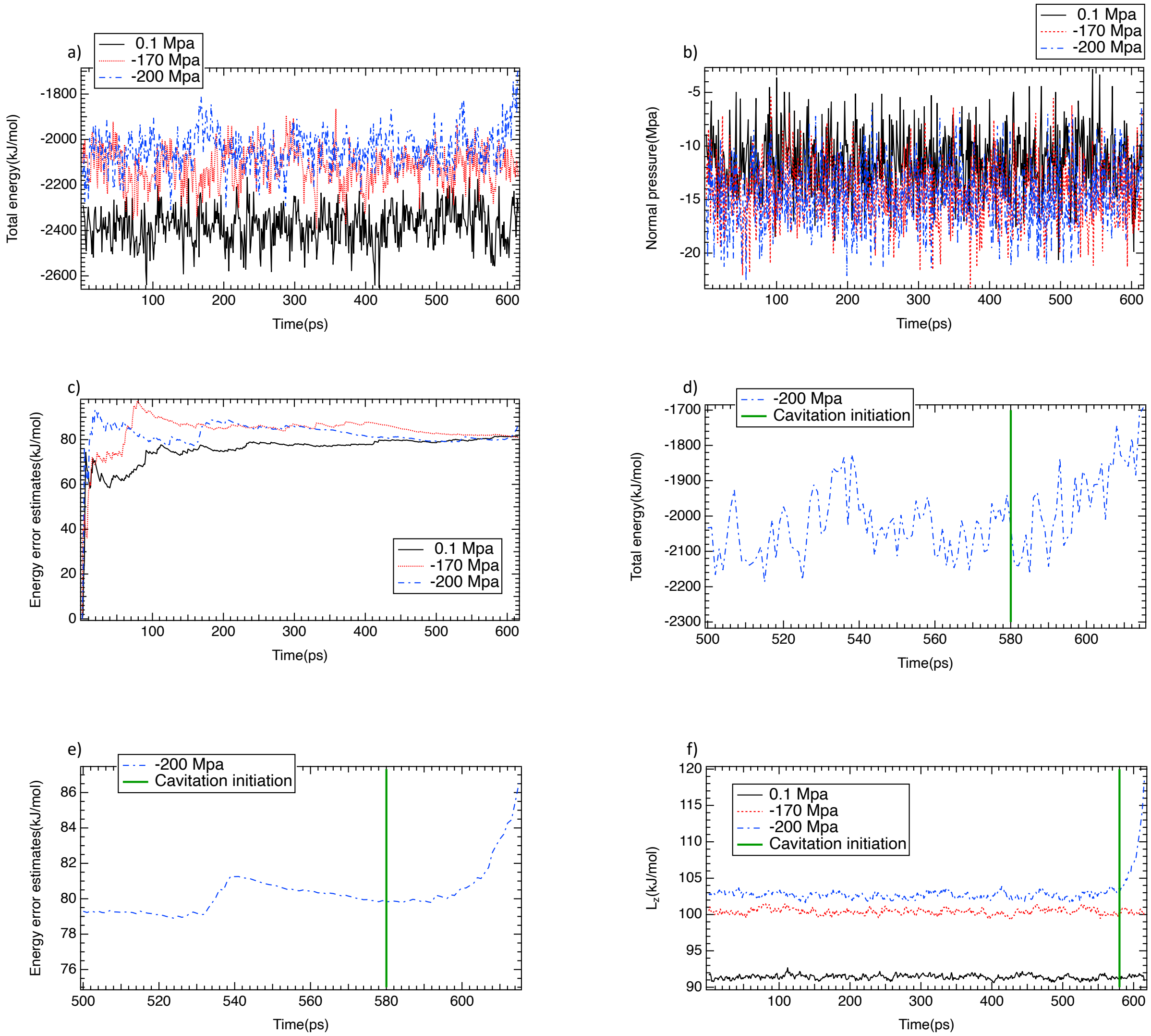
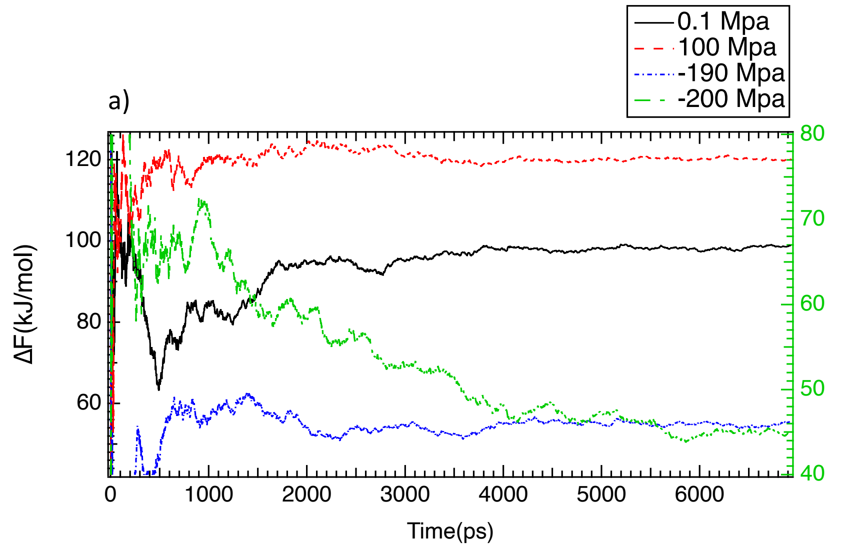


FIG. S4: (a) Total energy, (b) normal pressure, and (c) total-energy error estimates during the first 600 ps preceding cavitation at -200 MPa, -170 MPa, and 0.1 MPa at 280 K. (d) Total energy and (e) corresponding error estimates over 500–600 ps at -200 MPa and 280 K, highlighting fluctuations immediately prior to cavitation. (f) Simulation box length along the z direction as a function of time for the three representative pressures (-200 MPa, -170 MPa, and 0.1 MPa) at 280 K.



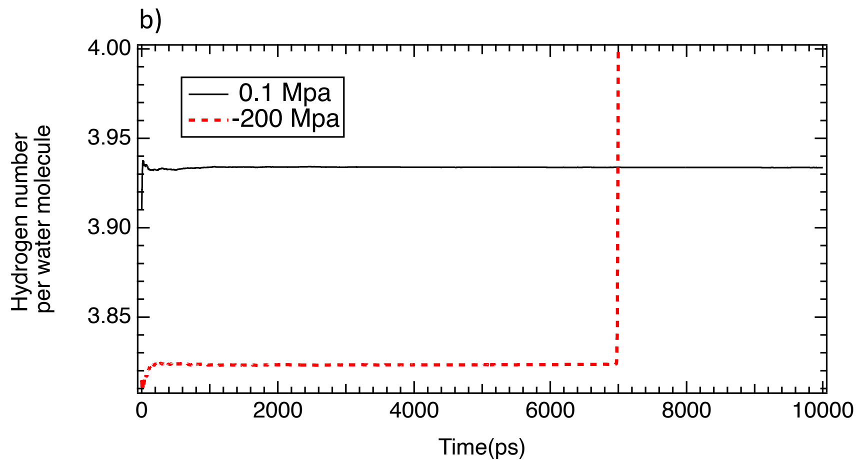


FIG. S5:(a) Time evolution of the interfacial free-energy difference for four representative pressures: -200 MPa (right axis), -190 MPa (left axis), -170 MPa (left axis), and 0.1 MPa (left axis) at 280 K. (b) Time evolution of the average number of hydrogen bonds per water molecule at 0.1 MPa and -200 MPa at 280 K.

REFERENCES

[1]A. Luzar and D. Chandler, Phys. Rev. Lett. **76**, 928 (1996).

[2]R. Renou, A. Szymczyk, and A. Ghoufi, J. Chem. Phys. **140**, 044704 (2014).